Supply chain Segmentation Analysis

DETAILED PROJECT REPORT

Shubham Tembhurne

PROJECT DETAIL

Project Title	US Pollution Analysis
Technology	Business Intelligence
Domain	Supply Chain Management
Project Difficulty level	Advance
Programming Language Used	None
Tools Used	Power Bi, Quadratric Ai

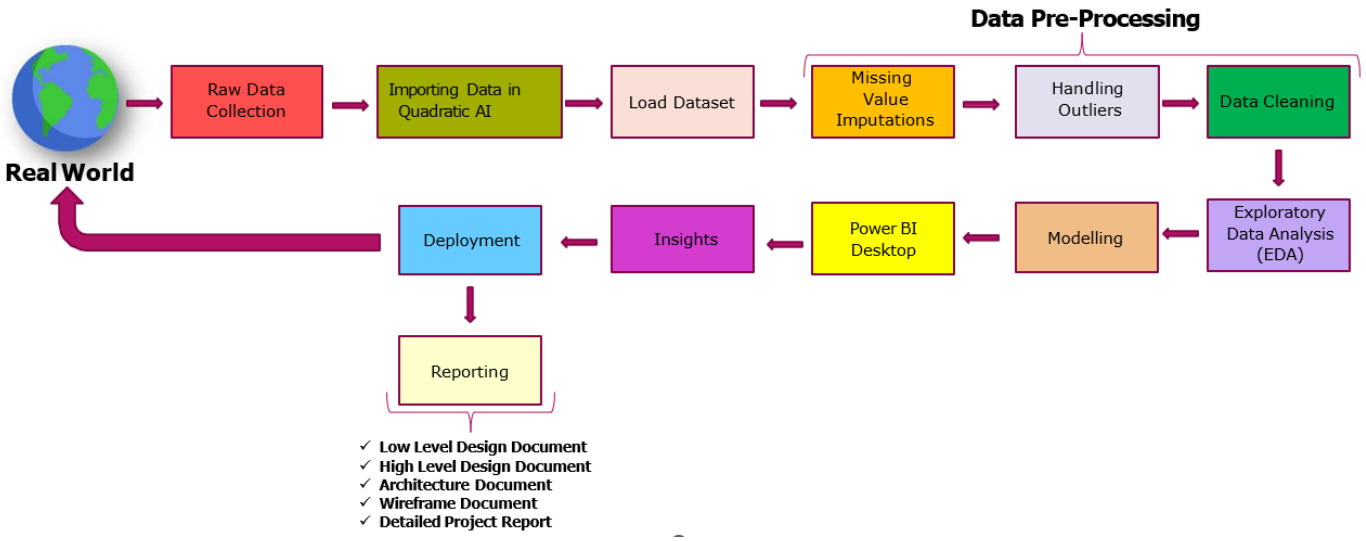
OBJECTIVE

- AtliQ Mart is a growing FMCG (fast moving consumer Good) manufacturer headquartered in Gujarat, India. It is currently operational in three cities Vadodara, Surat and Ahmedabad. They want to expand to other metro/tier1 cities in the next 2 years.

PROBLEM STATEMENT

- AtliQ Mart is currently facing a problem where a few key customers did not extend the annual contract due to service issues. It is speculated that some of the essential products were either not delivered on time or not delivered in full over a continued period, which could have resulted in bad customer service. Management wants to fix this issue before expanding to other cities and requested their supply chain analytics team to track the 'On time' and 'In Full' delivery service level for all the customers on a daily basis so that they can respond swiftly to these issues. The Supply Chain team decided to use a standard approach to measure the service level in which they will measure 'on-time delivery (OT) %', 'In-full delivery (IF) %' and On Time in full (OTIF) % of the customer orders on a daily basis against the target service level set for each customer.

Architecture



DATASET INFORMATION

- Line Fill Rate is an important metric for the supply planning team to understand how many lines they shipped out of the total lines ordered. This metric does not consider the delivery time of the order.
- Volume fill rate or case fill rate is a similar metric useful for the supply planning team to understand the total quantity they are able to ship for a customer per order or for a given period of time.
- On Time delivery %: OTD is a calculation of the number of shipments delivered on time to the customer in relation to the total number of orders shipped
- Full delivery %: This measure is measured at the order level. It determines if an order is delivered in full as per the requested quantity by the customer.
- OTIF%: OTIF or On-Time In-Full is a KPI used for measuring how many orders were delivered on time and in full. It helps to assess whether the business was able to deliver every item in the order on or before the expected date of delivery. This metric is mainly used as a delivery KPI, although it can also be applied throughout the supply chain.

This project contains these datasets:

- dim_customers.csv
- dim_products.csv
- dim_date
- dim_targets_orders
- fact_order_lines.csv
- fact_orders_aggregate.csv

Column Description for fact_order_lines: This table contains all information about orders and each item inside the orders.

1. order_id: Unique ID for each order the customer placed
2. order_placement_date: It is the date when the customer placed the

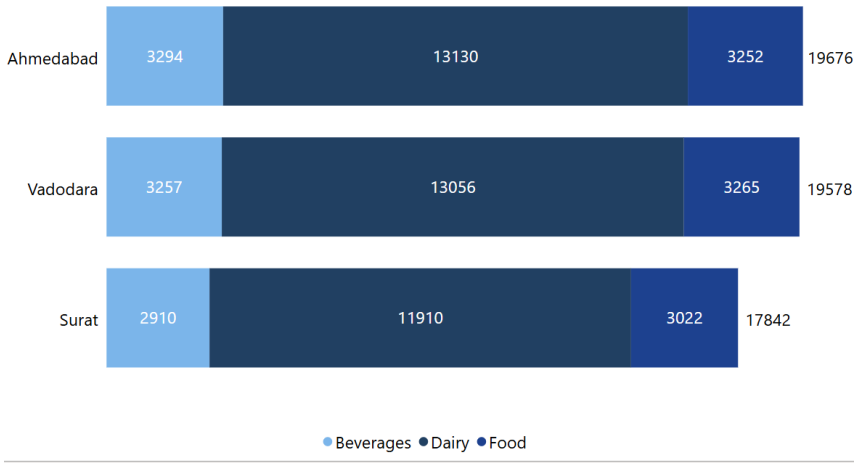
order
3. customer_id: Unique ID that is given to each of the customers
4. product_id: Unique ID that is given to each of the products
5. order_qty: It is the number of products requested by the customer to be delivered
6. agreed_delivery_date: It is the date agreed between the customer and AtliQ Mart to deliver the products
7. actual_delivery_date: It is the actual date AtliQ Mart delivered the product to the customer
8. delivered_qty: It is the number of products that are actually delivered to the customer

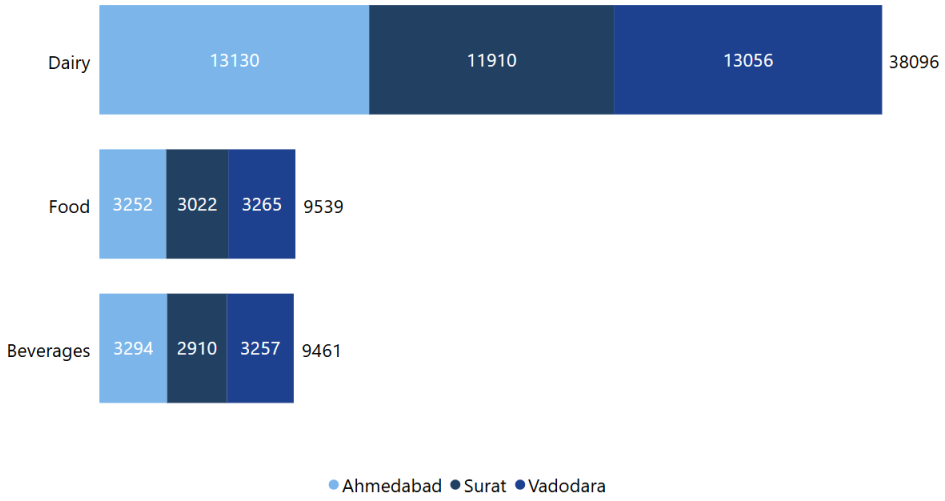
Column Description for fact_orders_aggregate: This table contains information about OnTime, InFull and OnTime Infull information aggregated at the order level per customer

1. order_id: Unique ID for each order the customer placed
2. customer_id: Unique ID that is given to each of the customers
3. order_placement_date: It is the date when the customer placed the order
4. on_time: '1' denotes the order is delivered on time. '0' denotes the order is not delivered on time.
5. in_full: '1' denotes the order is delivered in full quantity. '0' denotes the order is not delivered in full quantity.
6: otif: '1' denotes the order is delivered both on time and in full quantity. '0' denotes the order is either not delivered on time or not in full quantity

INSIGHTS

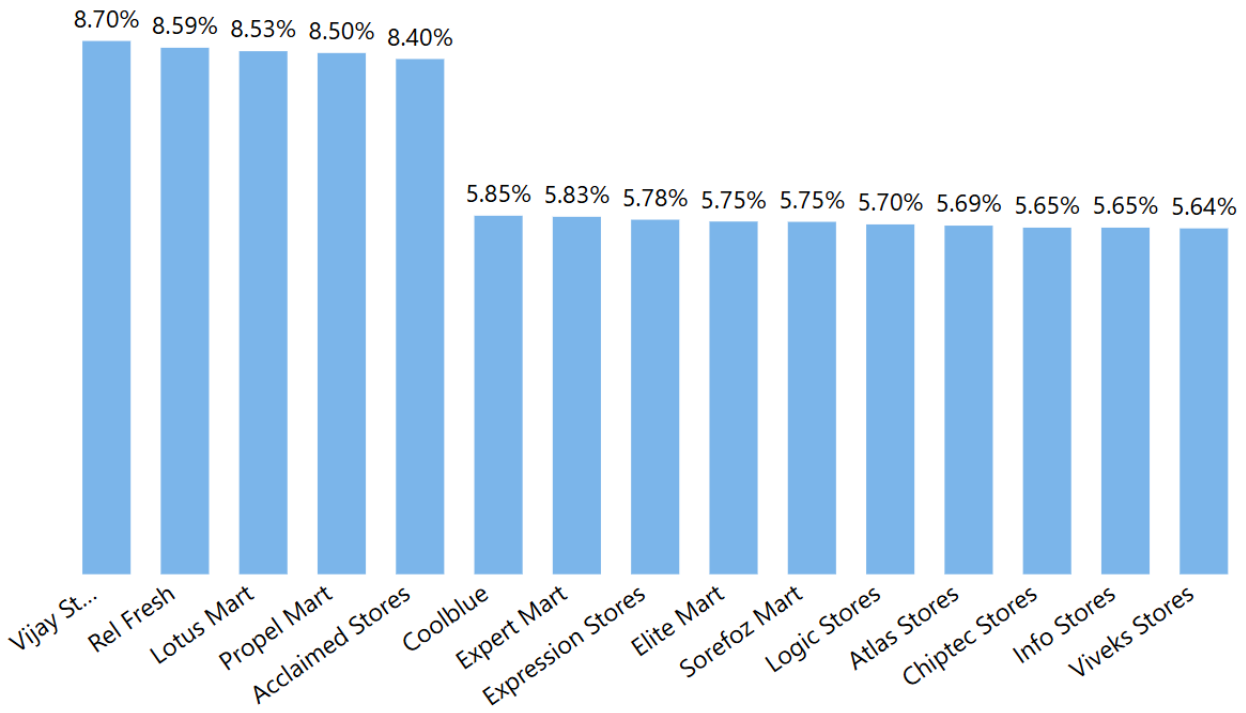
1. What is the city level performance and product at order level?





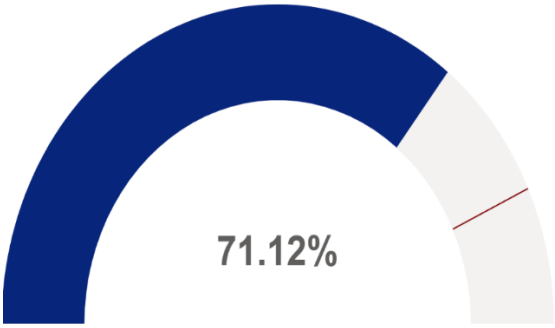
- The Ahmedabad city has the higher level of volume with total number of 19676 orders.
- Ahmedabad is the only City Which Contributes most of the orders.
- The Surat City has the lowest level of Performance with the low order of 17842 orders as compare to other cities.
- The Dairy product Generate the most of the orders almost 38096.
- Food and Beverages are performing the worst nearly fulling 9000 Orders.
- The Dairy product is mostly order in all the cities its high in demand.

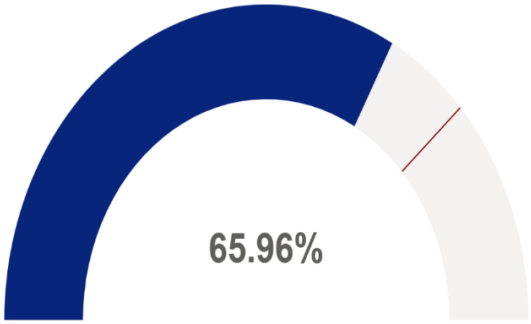
2. Which Stores contribute the highest number of orders?

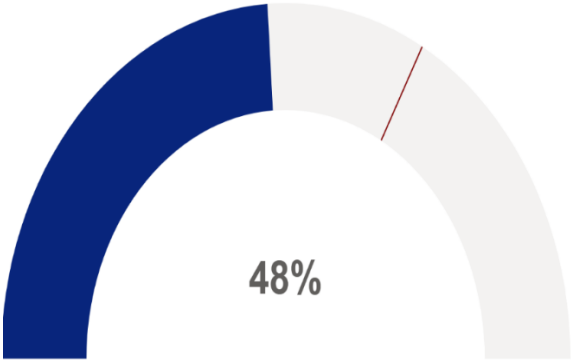


- The Vijay store, Rel fresh, Lotus Mart, Propel Mart, Acclaimed Stores Contribute the highest percentage of order Overall 8.40% to 8.70%.
- Coolblue, Expert Mart, Expression Stores, Elite Mart, Sorefoz Mart, Logic Stores, Atlas Stores, Info Stores, Vivek's Stores Contribute the lowest percentage of order overall 5.85% to 5.64%
- The demand is Concentrated only in few stores which are Contributing most of the orders stores like Vijay Store and Rel Fresh.

3. What is the Order Target at city level?



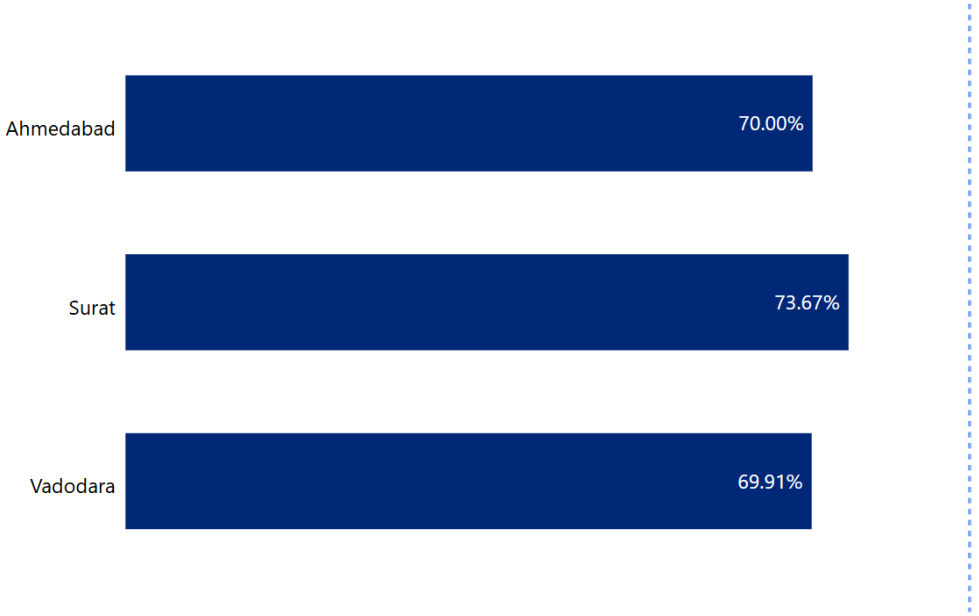




- The overall on time Delivery performance (OT%) is 71.12% unable to achieved the target which is 86%, We are facing -15% of the delivery gap.
- The overall in full Delivery performance (IF%) is 65.96% unable to achieved the target which is 77%, We are facing -11% of the delivery gap.
- The overall in on time In Full Delivery performance (Onif%) is 48% unable to achieved the target which is 66%, We are facing -18% of the delivery gap.

Ahmedabad
48%
Surat
51%
Vadodara
45%

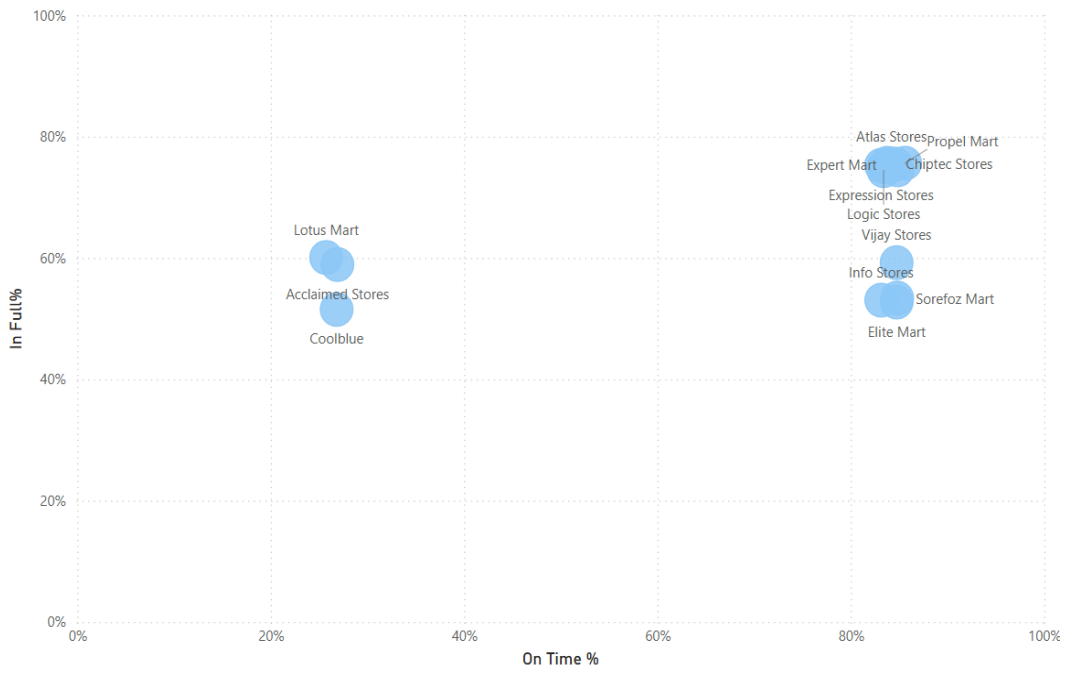
Ahmedabad
67.56%
Surat
66.69%
Vadodara
63.69%



- Surat is performing better than other cities with 73.67% of on time delivery.
- Vadodara is performing the worse than other cities with 63.69% of in full delivery.
- Surat is performing best in the on time in full deliveries 51% and Vadodara is the lowest performance city overall.
- Almost 30% of the deliveries are late.

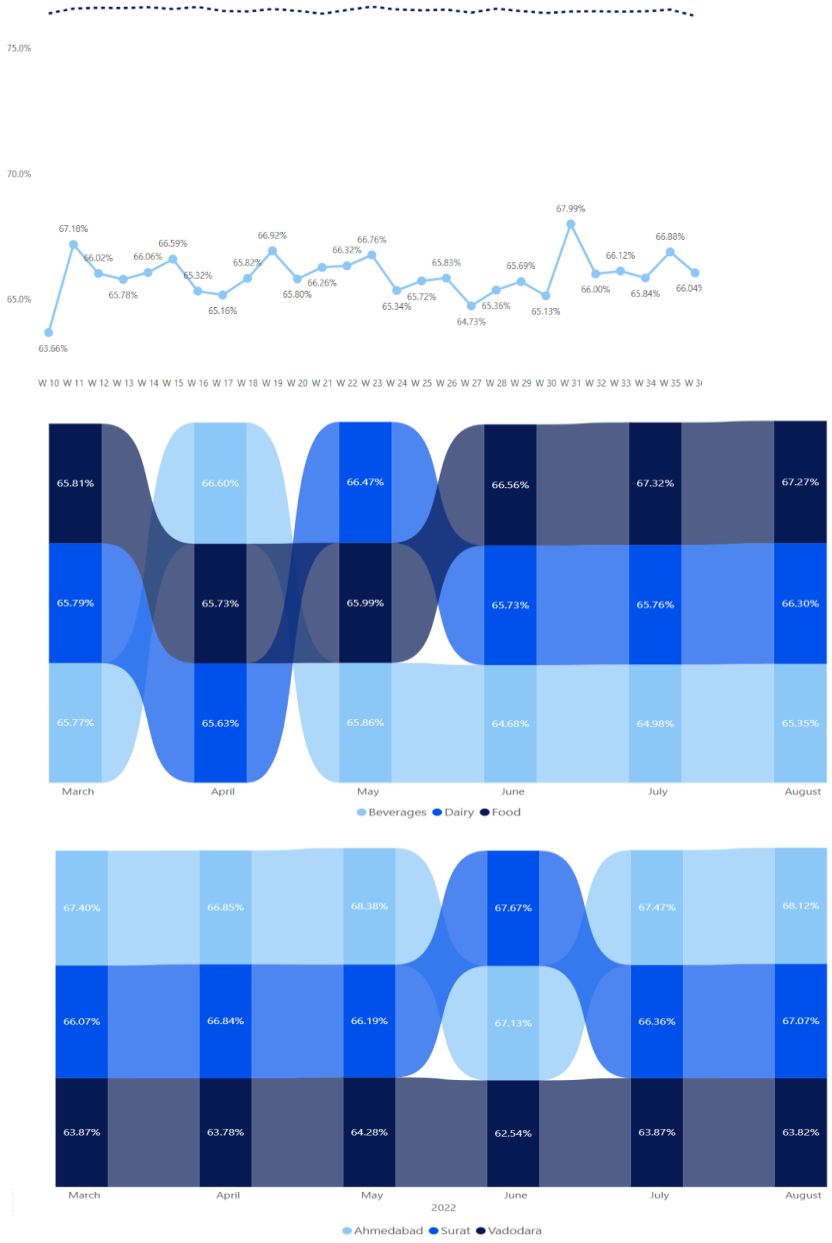
4.What is the service level analysis of the customers?

Metric	Acclaimed Stores	Atlas Stores	Chiptec Stores	Coolblue	Elite Mart	Expert Mart	Expression Stores	Info Stores	Logic Stores	Lotus Mart	Propel Mart	Rel Fresh	Sorefoz Mart	Vijay Stores	Viveks Stores
IF%	58.93%	75.48%	75.61%	51.53%	52.74%	75.48%	75.28%	53.05%	74.39%	60.08%	75.62%	74.54%	53.40%	59.23%	75.06%
OT%	26.89%	84.20%	83.76%	26.81%	84.74%	84.62%	83.13%	83.14%	83.39%	25.73%	85.62%	84.84%	84.79%	84.73%	83.42%
OTIF%	15.24%	63.80%	62.97%	13.75%	44.88%	63.61%	62.31%	43.48%	61.96%	16.06%	65.05%	63.04%	45.60%	49.61%	63.04%

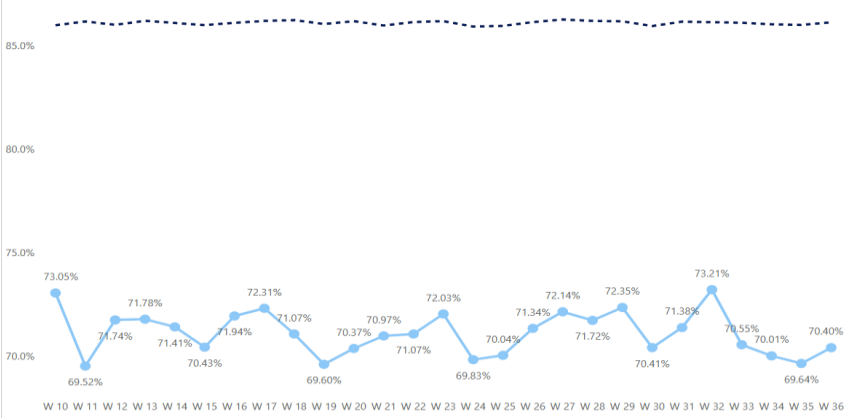


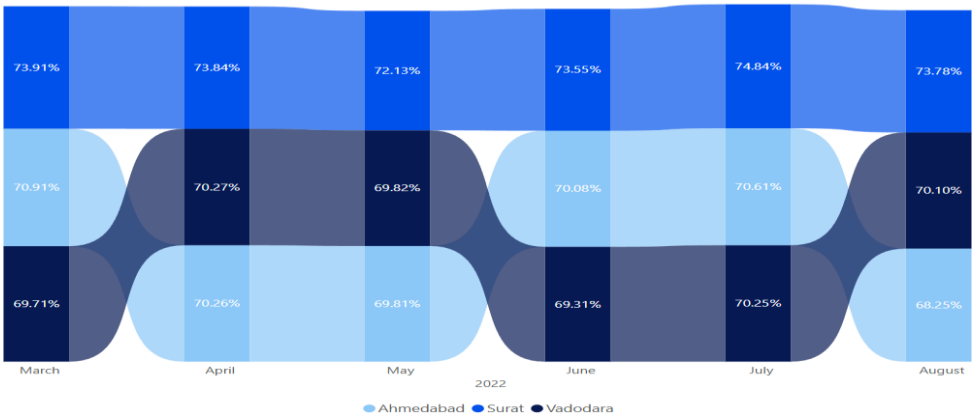
- Propel Mart customer has the highest on time in full performance almost 65 % of the deliveries.
- Locus Mart, Acclaimed Stores, Coolblue are not performing well because there Otif delivery rate percentage is lower than 20%
- Propel Mart customer is the best serve in the supply chain with 85% of the on-time delivery rate.
- Coolblue customer experience the worst service level with 13% of the on time in full delivery.
- Many customers Cluster around 80%-90% OT and 60%-75% IF

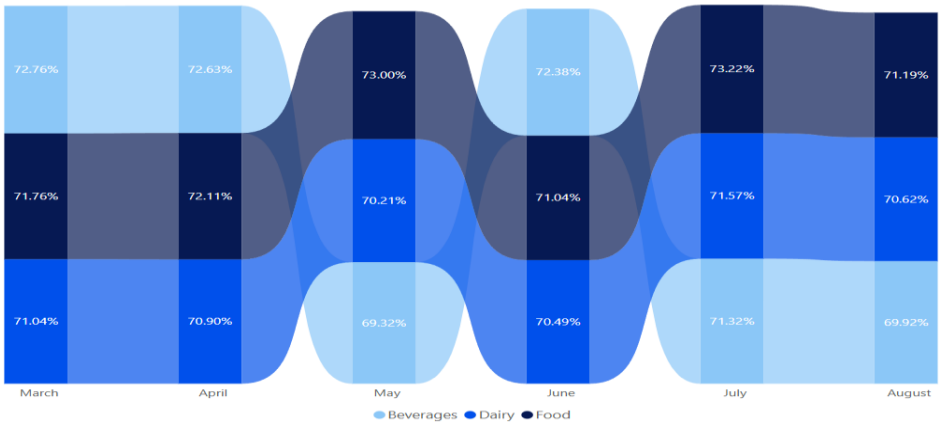
5.What is the Key trend of the Metrics (IF%, LIFR%, OT%, OTIF%, VOFR%)?



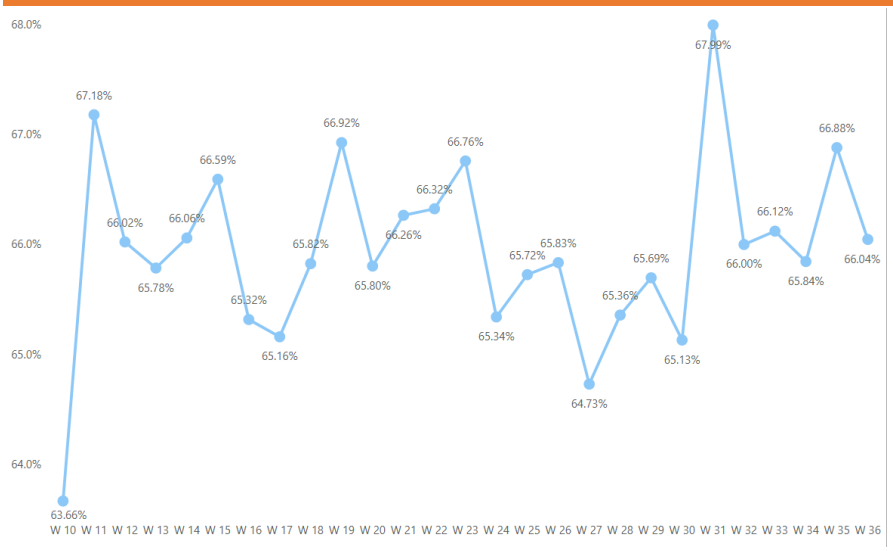
- The weekly IF% fluctuates around 65 to 67 % and the target line in 75 %
- The supply Chain consistently underperforms the target by 8 to 10%
- The IF% Values move between lowest 63% and highest 67-68% performance is stable but the consistency is below target.
- Vadodara consistently performs worst in fulfillment.
- Food Category has the lowest fulfillment rates.

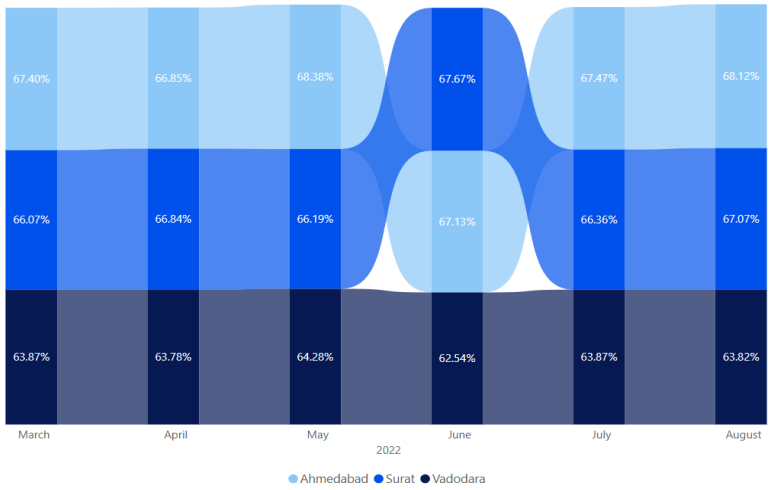


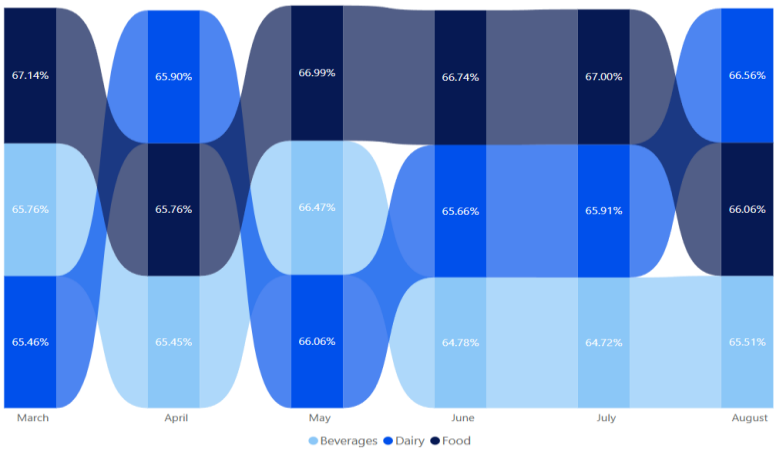




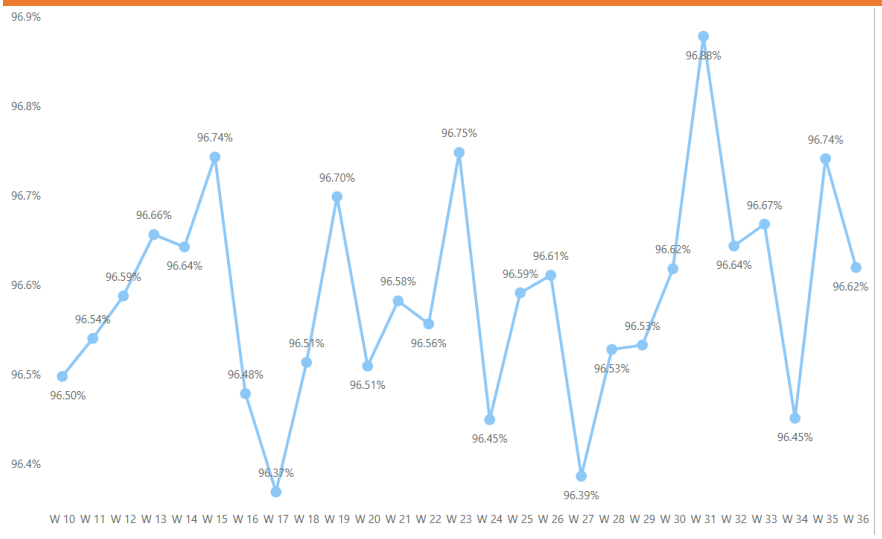
- The weekly OT% fluctuates around 69% to 73% the target is 83%.
- The company is completely missing the OT% target by 7% to 10%.
- The lowest week is week-11 69.5% and week-20 69.6%
- The highest week is week-17 72.3% and week-32 73.2%
- Vadodara has the lowest on time delivery performance 69 to 70%
- Food product has the most delivery delays.

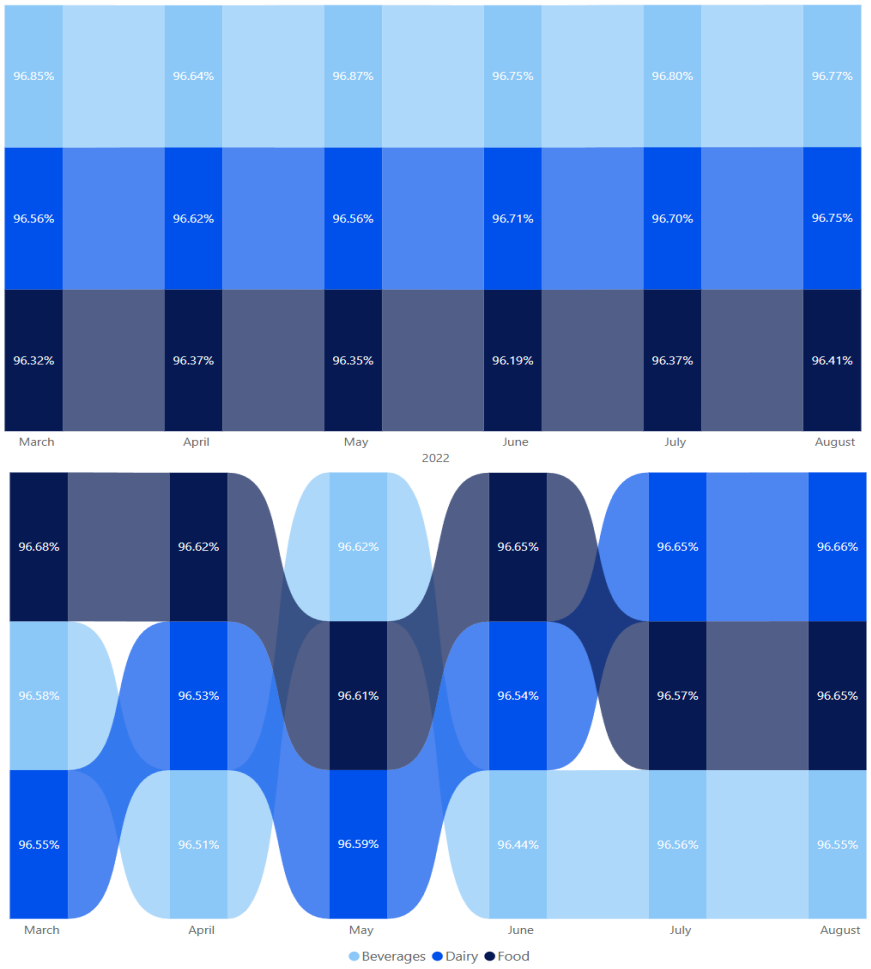






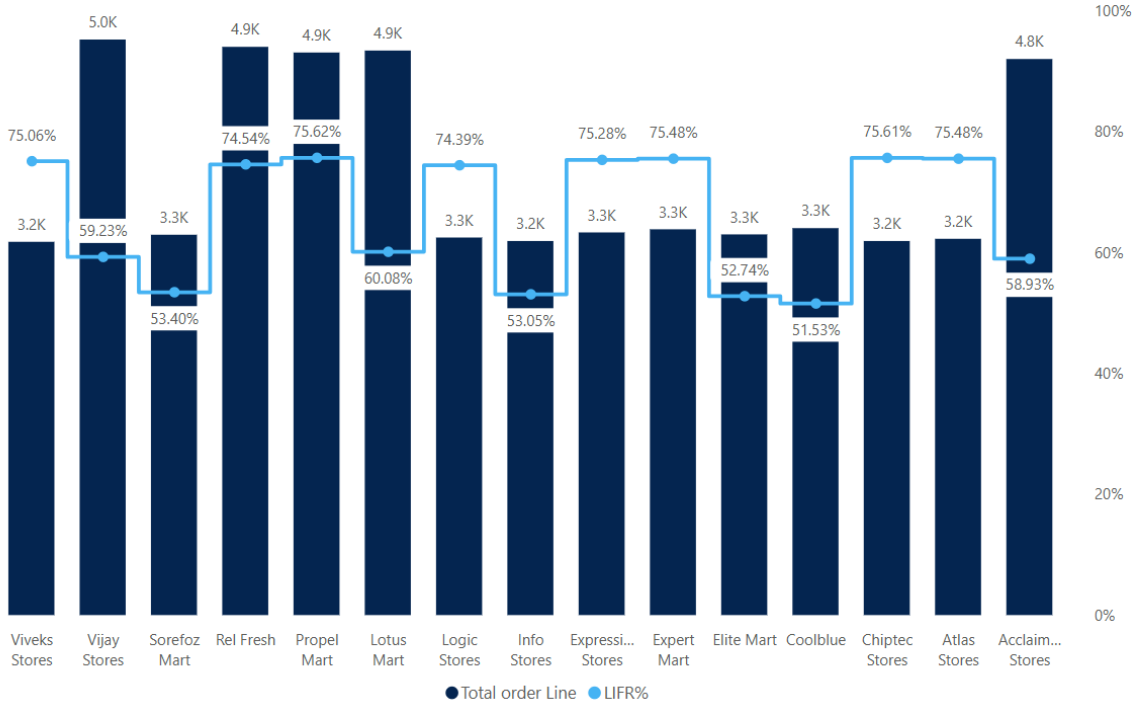
- The weekly LIRF fluctuates around 63% to 67%. Around 33-37% of the items experience shortages but no major improvement.
- Vadodara has the lowest line-item fulfillment performance around 64-65%
- Food products have the lowest fill rate around 64-65%





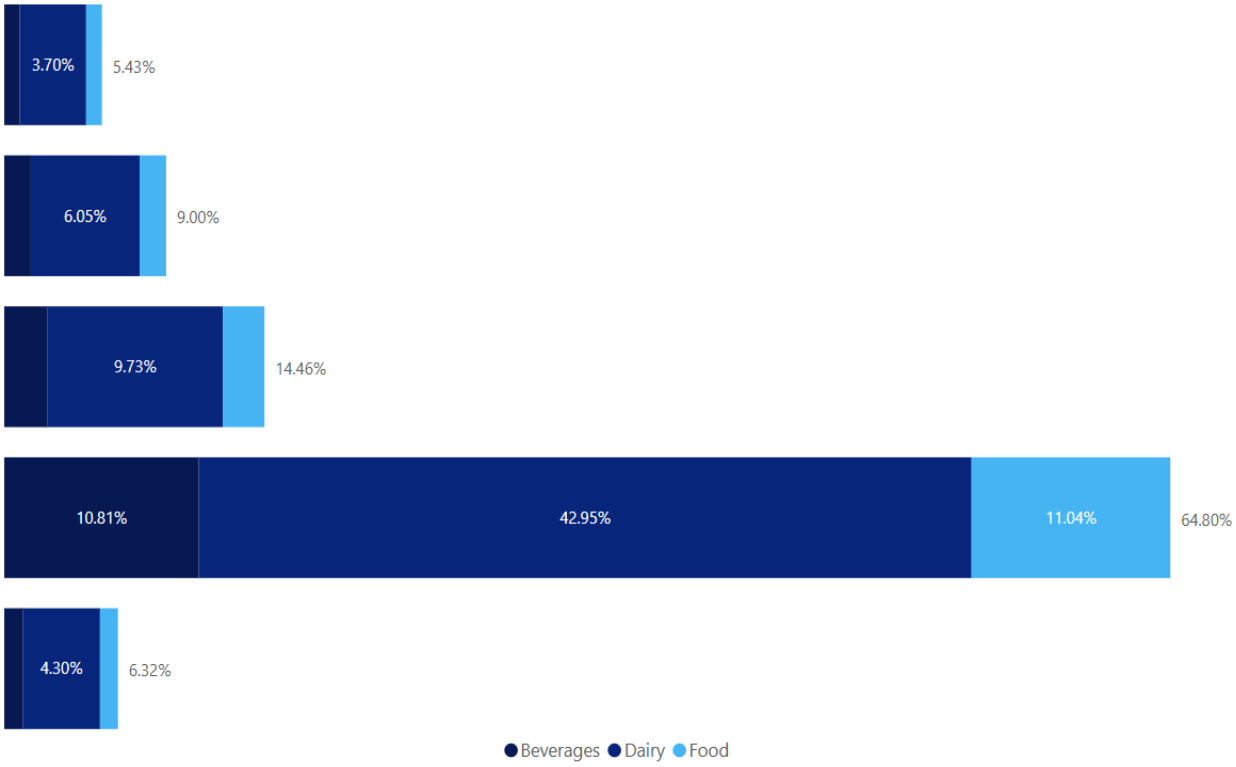
- The Weekly VOFR fluctuated around 96.3% to 96.9% shows very high Fulfillment performance.
- Vadodara has slightly lower fulfillment performance. (96.4%-96.4%)
- All product has very all high-volume fill rate, food category is slightly lower

6.How is the order line performance?



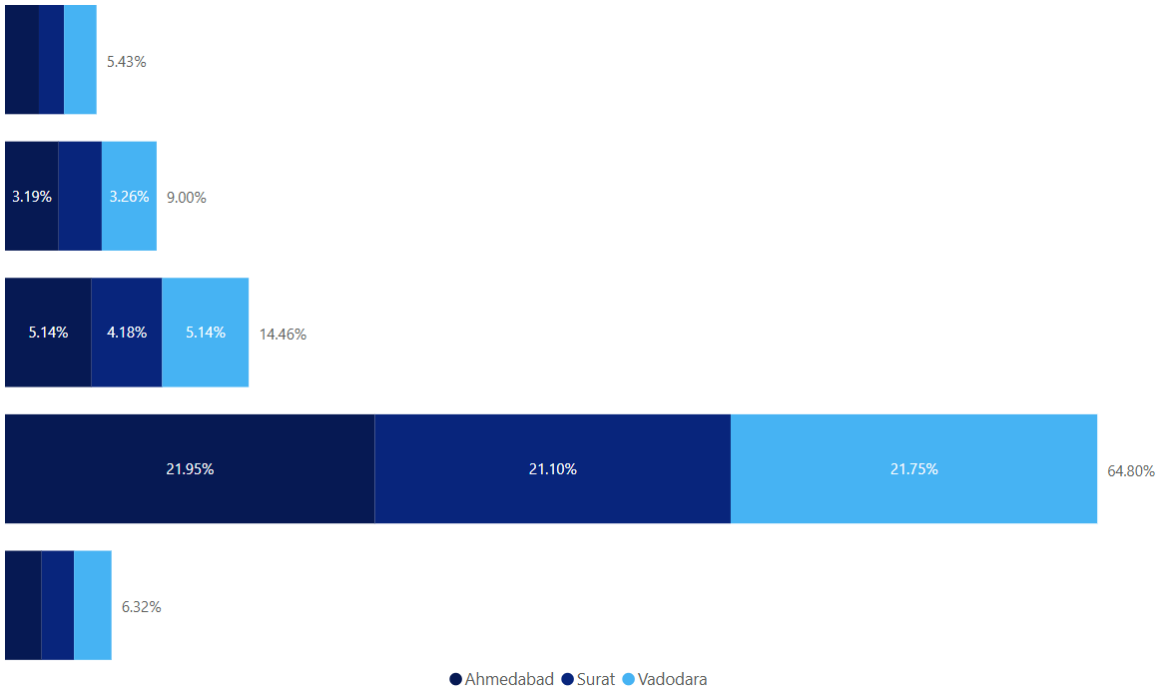
- The company processed 57,000 individual product line orders.
- 1.27M units the total order quality delivered across all orders.
- Only 38k order lines were delivered completely.
- 65.96% of the order line were fulfilled completely and 34% of orders had missing items
- Customers Vijay stores, Rel Fresh, Propel Mart, Acclaimed Stores this are high order volume customers around 4k to 5k orders.
- Customers Elite Mart (52%), Coolblue (51%), Info Stores (53%) with low LIFR% rate.
- Customers with high LIFR% equal to 75% chiptec stores, Atlas Stores, REL Fresh, Propel Mart
- One Third of the order line is incomplete

7.What is the lead line time Analysis?

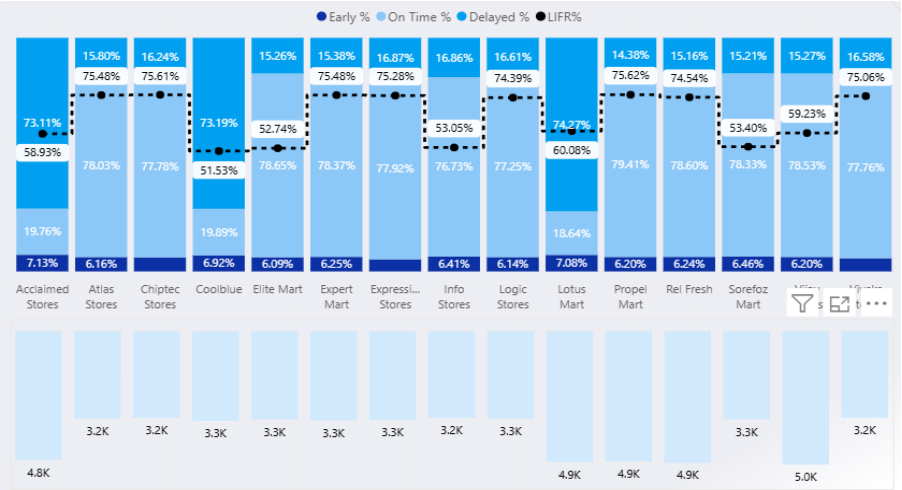


Delayed Days	Beverages	Dairy	Food	Total
-1	584	2454	569	**3607**
0	6171	24521	6306	**36998**
1	1377	5557	1321	**8255**
2	843	3452	841	**5136**
3	486	2112	502	**3100**
Total	**9461**	**38096**	**9539**	**57096**

Delayed Days	Ahmedabad	Surat	Vadodara	Total
-1	1243	1094	1270	**3607**
0	12531	12050	12417	**36998**
1	2934	2389	2932	**8255**
2	1819	1458	1859	**5136**
3	1149	851	1100	**3100**
Total	**19676**	**17842**	**19578**	**57096**



- Around 8255 orders had been one day delayed late
- Approximately 36,998 orders delivered on time
- Around 5,136 orders delayed by 2 days
- Around 3,100 orders delayed by 3 days
- Most on time deliveries come from Dairy product around 42.95% of contribution
- 14.6% deliveries are one day late is the biggest operational issue
- Furthermore ,2 days is 9% and 3 days is 5.43%
- All the cities are performing well with 21% of the delivery rate
- Food and beverages volumes are smaller so their impact is lower



- Most of the customers have 75% to 79% on time deliveries.
- Atlas Stores has Shown 78.03 % of on time delivery which is stable
- Elite Mart has Shown 78.65 % of on time delivery which is stable
- Propel Mart has Shown 79.41% of on time delivery which is stable
- Delayed deliveries are mostly round 14% to 16%
- Acclaimed Stores, Coolblue, Lotus Mart have the highest delayed deliveries around 73% to 74%
- Early deliveries around 6% to 7%
- LIFR% show low for customer Coolblue, Infostore, Sorefoz Mart and Vijay Store, Acclaimed stores around 51% to 59%
- Lotus Mart, Acclaimed Stores, Propel Mart, Rel Fresh, Vijay Stores have the large volume of orders around 4k to 5k.

8.How are the products are performing?

category	product_name	LIFR%	VOFR%	LIFR% by Date	VOFR% by Date
⊟ **Beverages**	AM Tea 100	65.90%	96.63%		
	AM Tea 250	65.34%	96.52%		
	AM Tea 500	65.12%	96.48%		
⊟ **Dairy**	AM Butter 100	66.17%	96.62%		
	AM Butter 250	64.25%	96.38%		
	AM Butter 500	65.64%	96.59%		
	AM Curd 100	66.25%	96.58%		
	AM Curd 250	66.30%	96.61%		
	AM Curd 50	65.57%	96.58%		
	AM Ghee 100	65.96%	96.57%		
	AM Ghee 150	66.46%	96.68%		
	AM Ghee 250	65.50%	96.53%		
	AM Milk 100	66.46%	96.63%		
	AM Milk 250	65.63%	96.62%		
	AM Milk 500	66.85%	96.63%		
⊟ **Food**	AM Biscuits 250	66.41%	96.64%		
	AM Biscuits 500	66.74%	96.62%		
	AM Biscuits 750	66.73%	96.63%		

- All the product has the 96% of the VOFR% delivery is almost in full order quantity.
- The LIFR% is totally low at most the average of 65% means the order line is not fulfilled completely

KEY PERFORMANCE INDICATOR(KPI)

1. In Full%, On Time%, Otif% compares to target
2. Otif% by City vs target
3. Metrics by customers, products
4. Otif% by top Customers vs target
5. LIFR and VOFR for top 10 Product by order volume
6. LIFR and VOFR trends over time for top 5 products
7. LIFR and VOFR trends over time for Products category
8. Weekly order Quantity for top 5 Cities
9. Regional Performance summary
10. Delivery delay analysis

CONCLUSION

- Ahmedabad is the only City Which Contributes most of the orders.
- The Dairy product is mostly order in all the cities its high in demand.
- Beverages have the lowest order share
- The Vijay store, Rel fresh, Lotus Mart, Propel Mart, Acclaimed Stores Contribute the highest percentage of order Overall 8.40% to 8.70%.
- The demand is Concentrated only in few stores which are Contributing most of the orders stores like Vijay Store and Rel Fresh.
- Surat performs slightly better in on-time deliveries.
- Vadodara is the lowest performing city overall.
- OTIF% = 48%, This means more than half the orders fail service standards.
- IF% = 65.96%, Many orders are shipped with missing items.
- OT% = 71.12%, Almost 30% of deliveries are late
- Atlas Stores, Chiptec Stores, Propel Mart, Expert Mart. These customers have strong supply chain service levels.
- Many customers cluster around 80-90% OT and 60-75% IF
- The supply chain consistently underperforms the target by 8-10%
- Weekly IF% Trend Lowest ≈ 63% Highest ≈ 67-68% performance is stable but consistently below target.
- Vadodara consistently performs worst in fulfillment.
- Food category has the lowest fulfillment rate.

- Weekly OT% fluctuates between approximately 69% to 73%
- The company consistently misses the delivery target by 6-9%.
- Vadodara has the lowest On-Time delivery performance.
- Food products have the most delivery delays.
- Weekly LIFR fluctuates around 63% – 67%
- Vadodara has the lowest line-item fulfillment performance.
- Food products have the lowest fill rate.
- Weekly VOFR fluctuates around: 96.3% – 96.9%
- All product categories maintain very high-volume fulfillment
- 38K order lines were delivered completely
- Vijay Stores, Rel Fresh, Propel Mart, Acclaimed Stores place the highest number of orders.
- Customers Elite Mart (52%), Coolblue (51%), Info Stores (53%) with low LIFR% rate
- One-third of order lines are incomplete
- Most orders are delivered on time, but a significant portion experiences 1-day delays.
- City-level analysis shows consistent performance across Ahmedabad, Surat, and Vadodara, indicating that delays are likely caused by operational inefficiencies
- The biggest operational issue is delayed deliveries, not early or on-time ones.
- Even though on-time delivery is good, the delay rate 15% average is still significant.
- The biggest supply chain problem is not quantity fulfillment but product availability.

- Customers are receiving most of the volume, but not the exact products they ordered.

STRATEGIC RECOMMENDATION:

- Increase Dairy inventory allocation.
- Monitor low-performing stores or categories.
- Improve logistics in cities with lower OTIF performance.
- Improve inventory availability to increase IF%.
- Optimize delivery routes to improve OT%.
- Focus improvement initiatives in Vadodara.
- Improve service for low-performing customers
- Investigate transportation delays for stores with low OT% but good IF%.
- Improve inventory allocation for stores with low IF%.
- Investigate operational issues during low-performance weeks.
- Improve Food category supply chain and Reduce shortages and improve inventory availability.
- Focus on Vadodara distribution Improve fulfillment in this city.
- Investigate Food category shortages It shows slightly lower fulfillment.
- Focus on low-performing customers Investigate why customers like Coolblue and Elite Mart receive incomplete orders.
- Prioritize high-volume customers Ensure consistent service levels for major customers.
- Optimize transport scheduling Optimize transport scheduling
- Focus on high-volume category Improve Dairy logistics efficiency.
- Improve inventory availability Prevent incomplete deliveries that reduce LIFR.
- Focus on low LIFR products with lowest LIFR should be investigated first.

Q & A

Q1) What's the source of data?

Ans)
https://drive.google.com/drive/folders/
1AkVENpnXau9MdW87bVmLtuBFap8R2
i6m

Q2) What was the type of data?

Ans) The data was a combination of numerical and Categorical values.

Q3) What's the complete flow you followed in this Project?

Ans) Refer slide 5th for better Understanding

Q4) What techniques were you using for data?

Ans) -Using Quadratic Ai for cleaning and
merging the dataset Help is to reduce manual
work and increase efficiency

- Using Prompt for the EDA with the help of
power BI

-Removing unwanted attributes

-Visualizing relation of independent variables with each other and output variables

-Removing outliers

-Cleaning data and imputing if null values are present.

-Converting Numerical data into Categorical values.

Q6) What were the libraries that you
used in Python?

Ans) I haven't used any kind of programming language

THANK YOU